

TUGAS 5

SISTEM OPERASI



Nama : Aan Maulana Sampe
Stambuk : 13020230081
Dosen : Syahrul Mubarak Abdullah, M.Kom., MTA.

PROGRAM STUDI TEKNIK INFORMATIKA
FAKULTAS ILMU KOMPUTER
UNIVERSITAS MUSLIM INDONESIA
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Tugas 5

Membuat 2 contoh masing-masing untuk perhitungan algoritma Priority Scheduling dan algoritma Round Robin

Jawaban:

Contoh 1 Priority Scheduling

P	BT	Priority
P ₁	5	3
P ₂	15	2
P ₃	12	1
P ₄	23	2
P ₅	99	4

Gant chart

P3	P2	P4	P1	P5	
0	12	27	40	45	144

P	WT
P ₁	40
P ₂	12
P ₃	0
P ₄	27
P ₅	45

$$= \frac{40 + 12 + 0 + 27 + 45}{5}$$

$$= \frac{124}{8} = 24.8 \text{ ms}$$

Contoh 2 Priority Scheduling

P	BT	Priority
P ₁	10	1
P ₂	12	2
P ₃	15	3
P ₄	20	5
P ₅	25	4

Gant chart

P ₁	P ₂	P ₃	P ₅	P ₄	
0	10	22	37	62	82

No

Date

P	WT	
P ₁	0	$= \frac{0+10+22+62+37}{5}$
P ₂	10	
P ₃	22	
P ₄	62	
P ₅	37	$= \frac{131}{5} = 26.2 \text{ ms}$

Contoh 1 algoritma Round Robin

P	BT	Quantum Time (QT) = 5																			
P ₁	9/4/-	Gantt chart <table border="1"><tr><td>P₁</td><td>P₂</td><td>P₃</td><td>P₄</td><td>P₅</td><td>P₁</td><td>P₂</td><td>P₃</td><td>P₅</td></tr><tr><td>0</td><td>5</td><td>10</td><td>15</td><td>20</td><td>25</td><td>29</td><td>34</td><td>37</td><td>42</td></tr></table>	P ₁	P ₂	P ₃	P ₄	P ₅	P ₁	P ₂	P ₃	P ₅	0	5	10	15	20	25	29	34	37	42
P ₁	P ₂		P ₃	P ₄	P ₅	P ₁	P ₂	P ₃	P ₅												
0	5		10	15	20	25	29	34	37	42											
P ₂	10/5/-																				
P ₃	8/3/-																				
P ₄	5/-																				
P ₅	10/5/2																				

Average Waiting Time

P	WT	
P ₁	0 + (25 - 5) = 20	$= \frac{20+24+29+15+32}{5}$
P ₂	5 + (29 - 10) = 24	
P ₃	10 + (34 - 15) = 29	
P ₄	15	
P ₅	20 + (37 - 25) = 32	
		$= \frac{120}{5} = 24 \text{ ms}$

Contoh 2 Algoritma Round Robin

P	BT
P ₁	10/4/-
P ₂	11/5/-
P ₃	9/3/-
P ₄	6/-
P ₅	12/6/-

Quantum Time (QT) = 5

P ₁	P ₂	P ₃	P ₄	P ₅	P ₁	P ₂	P ₃	P ₅	
0	6	12	18	24	30	34	39	42	48

AWT

P	WT
P ₁	0 + (30 - 5) = 25
P ₂	6 + (34 - 12) = 22
P ₃	12 + (39 - 18) = 21
P ₄	18
P ₅	24 + (42 - 30) = 12

$$= \frac{25 + 22 + 21 + 18 + 12}{5}$$

$$= \frac{98}{5} = 19.6$$